

Title: Design, Model, and Test Advanced Communication and Sensing Electronics at Temperatures near 4 Kelvin

Sponsor: WSU Systems-on-Chip Lab, and Boeing

Abstract

Project significance: Boeing is seeking to have Washington State University support its cryogenic research. Boeing seeks to have multiple commercial off the shelf (COTS) parts modeled, designed, and tested in deep cryogenic environment (~ 4 Kelvin) with the performance of the parts characterized and its behavior studied for applications such as radar and satellite communications.

Tasks:

Boeing will provide an initial list of parts for selection starting with block-level specifications and gradually build towards a high-level application such as RADAR or Localization by the WSU Systems-on-Chip Lab. Upon a mutually agreed list, the team will then model and develop cryo-compatible printed-circuit boards under guidance of senior PhD students at the systems-on-chip lab. It is expected that there will be 3 or 4 different components selected, and equivalently integrated into the PCB in a stepwise approach.

Post the cryo-characterization, the team will understand any non-ideal (unexpected) behaviors that will happen due to deep cryogenic operation (below 20K) to 4K. The study of these behaviors is expected to lead to new knowledge and application in the RADAR / Wireless communication domains.

The team composition will benefit from being inter-disciplinary with students from different traits in EECS welcome to the proposed research and development effort. Ideally, the team members would have taken 300- and 400-level courses in:

1. Signal Processing
2. Digital Communications
3. Electromagnetics
4. Embedded Systems
5. Microelectronics

The selected member will also be working with both physical hardware (requiring bench testing in the electronics lab) as well as Matlab and C++/Python simulation software with interest in embedded systems, Altium (or alternative software for Printed Circuit Board (PCB)), Electromagnetic simulations (antenna and transmission line design). The team will validate the Cryoelectronics performance in room-temperature settings followed by a more comprehensive radio performance in deep cryogenic environments.

Industry mentorship: The project will have industry mentors from Boeing periodically with regular calls. One Hour weekly meeting with the mentor / co-mentors and collaborators is expected besides time commitment in the cryogenic lab under PI Gupta.

